Mariia Pavlenko

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WORK & LEADERSHIP EXPERIENCE

University of Oxford **Oxford, United Kingdom**
Master of Science in Physics and Philosophy Sep 2019 - Aug 2023

- **Grades:** 2.I (Upper Second Class). A-Levels: A*A*A*A* in Physics, Chemistry, Maths, Further Maths.
- **Relevant Coursework:** Machine Learning, Linear Algebra, Statistical Analysis, Advanced Quantum Physics.

WORK & LEADERSHIP EXPERIENCE

Flow Traders **London, United Kingdom**
Quantitative Researcher Feb 2024 - Current

- Built end-to-end machine learning pipelines for a range of trading strategies, including sourcing new data, developing models, deploying to production, and maintaining systems.
- Developed proprietary modular pricing engine with extensive data integration (raw price data, news, external data such as weather), fine-tuned models with pre-build features as the main pricing logic and a variety of consumption sources such as Grafana, custom made UI and more
- The product was rolled out to three initial users with focus on energy markets, which has led to over 80%+ less downtime, 7.2% improvement in price accuracy and a leap in usability, UX and UI
- Developed proprietary APIs using Python, integrated with low-latency systems (Kafka), various databases (ClickHouse, MariaDB), and cloud environments (GCP, AWS Lambda) on Linux infrastructure.

Entrepreneur First **London, United Kingdom**
Founder in residence Jun 2023 - March 2024

- Alumni of Europes first start-up accelerator focused on applications of technology across industries.
- Focused on three key projects: applications of zero knowledge proofs and homomorphic encryption within ML & data sharing, payment systems in open banking, and AI driven treatment plans within therapeutics.
- Developed an IOS-based app and onboarded 100 initial users, handled investor relations and product development.

Maven Securities **London, United Kingdom**
Trading Intern (return offer) Jun - Aug 2022

- Applied options theory and various market making strategies (such as delta neutral and mean reversion) in a simulated trading environment focused on trading Euro Stoxx.
- Created an auction monitoring web based application through a 4-week project using Python, HTML, CSS and JavaScript which is now in use on the trading floor.
- Worked in a team of traders in developing options trading strategies by analyzing the macroeconomic environments.

IMC Trading **Amsterdam, Netherlands**
Female Trading Program Sep - Oct 2021

- Developed an understanding of options theory and algorithmic trading through internal courses and mentorship schemes, working closely with experienced traders.
- Created options valuation and delta-neutral hedging algorithms using Python, which were then deployed on real market data in an internal coding competition in which I placed top 10.

Streater Research (equity research shop) **London, United Kingdom**
Equity Research Analyst Jul– Sep 2021

- Conducted industry analysis with a focus on the commodities and energy sectors.
- Presented my key findings on the current and future developments within the field in a 100-page report which explored various avenues for the energy transition and the key players behind it.
- Created a data scraping algorithm to analyze oil and gas prices on demand using Python and Excel VBA.

fair (fintech start-up) **Oxford, United Kingdom**
Co-founder Feb 2021 – Jan 2022

- Founded a fintech start-up aiming to become a financial education provider and ESG-focused EU brokerage.
- Built and led a team of 50+ web developers, marketers, and financial researchers using a scrum methodology.
- Won the UCL EFS start-up competition after entering the final round with 6 other teams, passing a three-day

bootcamp lead by industry professionals and pitching our business model to VC investors.

Amazon **London, United Kingdom**

Spring Technology Insight Apr 2021

- • Gained an insight into the field of software engineering, developing the relevant coding skills
- • Used HTML and SQL for front- and back-end projects which I then presented to my mentor in a form of a website.

SKILLS, ACTIVITIES, & LEADERSHIP

Programming Experience:

- Languages & Frameworks: Proficient in Python, JavaScript/TypeScript, and React. Intermediate in HTML, CSS, Solidity, and Rust
- Systems & Databases: Experienced with GCP and AWS-based systems, Linux environments, and various databases (ClickHouse, PostgreSQL, MariaDB)

Machine Learning:

- Applied neural networks and transformer models to image classification projects with the Oxford AI Society.
- Used reinforcement learning to optimize commercial product strategy at Oxford Strategy Group Digital.
- Finalist at AI Hack London 2024, focusing on innovative ML applications

Software Engineering:

- Amazon Spring Internship: Contributed to web development projects involving both front-end (HTML/CSS) and back-end (SQL) systems

Blockchain Development:

- Active developer and member of the Oxford Blockchain Society.
- Participated in 10+ international hackathons with a focus on DeFi, asset tokenization, ETF/fund management, and blockchain-based payment systems.
- Achieved 8+ finalist positions and bounty wins

Leadership & Community Engagement:

- Invited speaker and panellist at various blockchain and fintech events:
  - EthOxford: Panellist on fintech and technical startup development (500+ attendees).
  - Africa Blockchain Society: Speaker on scaling startups and fundraising (100+ attendees).
- Organizer of Avalanche London Summit 2025 (150+ attendees).
- Served as a technical mentor for full-stack development at multiple events